to wait far in excess of the age of the universe for the constituents to, of their own accord, regroup in their initial configuration. Nevertheless, as a point of principle, it is provocative to note that with endless patience and longevity, every spatially contained physical system will return to how it was initially configured.

11. You might wonder, then, why water ever turns into ice, since that results in the H_2O molecules becoming more ordered, that is, attaining lower, not higher, entropy. Well, the rough answer is that when liquid water turns into solid ice, it gives off energy to the environment (the opposite of what happens when ice melts, when it takes in energy from the environment), and that raises the environmental entropy. At low enough ambient temperatures, that is, below 0 degrees Celsius, the increase in environmental entropy exceeds the decrease in the water's entropy, so freezing becomes entropically favored. That's why ice forms in the cold of winter. Similarly, when ice cubes form in your refrigerator's freezer, their entropy goes down but the refrigerator itself pumps heat into the environment, and if that is taken account of, there is a total net increase of entropy. The more precise answer, for the mathematically inclined reader, is that spontaneous phenomena of the sort we're discussing are governed by what is known as *free energy.* Intuitively, free energy is that part of a system's energy that can be harnessed to do work. Mathematically, free energy, F, is defined by $F = U - TS$, where U stands for total energy, T stands for temperature, and S stands for entropy. A system will undergo a spontaneous change if that results in a decrease of its free energy. At low temperatures, the drop in U associated with liquid water turning into solid ice outweighs the decrease in S (outweighs the increase in $-TS$), and so will occur. At high temperatures (above 0 degrees Celsius), though, the change of ice to liquid water or gaseous steam is entropically favored (the increase in S outweighs changes to U) and so will occur.

12. For an early discussion of how a straightforward application of entropic reasoning would lead us to conclude that memories and historical records are not trustworthy accounts of the past, see C. F. von Weizsäcker in The Unity of Nature (New York: Farrar, Straus, and Giroux, 1980), 138–46, (originally published in *Annalen der Physik* 36